

Letter of Recommendation for Mr. Gaurav Santosh Gokhale

To Whom It May Concern,

It is our distinct pleasure to write this letter of recommendation for **Mr. Gaurav Santosh Gokhale**, who worked as a **C++ Software Developer** at **DCS Computing GmbH** from **January 2025 to July 2026** within our **Aspherix simulation software** team.

DCS Computing is a specialized developer and provider of advanced particle simulation software, recognized globally for our high-performance Discrete Element Method (DEM) platform, Aspherix, and our coupled CFD-DEM solutions.

During his time with our company, Gaurav made outstanding contributions to the development, optimization and maintenance of our high-performance discrete element simulation platform. He worked on performance-critical components of the codebase with a focus on GPU acceleration, parallel computing, and simulation reliability. His responsibilities included the implementation and optimization of heterogeneous computing backends, enhancement of the build and testing infrastructure, and the development of efficient and maintainable solutions for complex simulation workloads.

One of Gaurav's key strengths was his ability to identify performance bottlenecks and translate them into robust, maintainable engineering solutions. He made significant contributions to the optimization of our GPU execution pipeline, including a redesign of the particle-mesh neighbor-list algorithm. In addition, he reworked aspects of the GPU memory allocation and management strategy, reducing unnecessary allocations and CPU-GPU data movement overhead. These efforts collectively resulted in substantial improvements in simulation performance and scalability of Aspherix.

Beyond algorithmic optimization, Gaurav contributed to the ongoing modernization of the Aspherix codebase. By refactoring core components to achieve a cleaner separation between internal libraries and modules, improving maintainability and enabling more modular development. In doing so, he applied modern C++ design principles and language features to improve code quality and expressiveness while preserving the performance characteristics required of a high-performance simulation codebase. He also enhanced the CMake-based build infrastructure, streamlining development workflows, improving cross-platform compatibility, and extending support for heterogeneous computing backends including CUDA, HIP, and OpenMP. His work helped establish a more flexible and maintainable foundation for future development.

Gaurav was equally comfortable working on low-level performance engineering and complex debugging tasks. He successfully investigated and resolved challenging distributed-memory issues, including MPI-related calculation errors, segmentation faults, and invalid memory accesses, using advanced debugging techniques involving GDB, Python-GDB integration, and custom analysis tools. He consistently demonstrated strong analytical skills and a methodical approach to problem solving, allowing him to tackle technically demanding problems with a high degree of independence and reliability.



Gaurav performed all tasks assigned to him to our complete satisfaction. In addition to his technical expertise, he was a dependable, motivated, and a collaborative member of our engineering team. He approached new challenges with curiosity and professionalism and was always willing to take ownership of difficult problems. His ability to combine deep algorithmic understanding with sound software engineering practices enabled him to deliver solutions that improved not only performance, but also the maintainability and long-term evolution of the codebase.

Gaurav is leaving our company due to an internal operational restructuring and a subsequent realignment of tasks. We deeply regret losing him, as he was a highly valued employee both professionally and personally.

We are fully confident that Gaurav will be an extremely valuable asset to any organization seeking a highly capable software engineer with expertise in high-performance computing, GPU programming, modern C++, and simulation software development. We thank him for his excellent work and commitment, and we wish him every success and all the very best in his future professional and personal endeavors.

Linz, 17.06.2026

Sincerely,

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Dr. Christoph Kloss
Managing Director
DCS Computing GmbH

